

TASK-3 REPORT

Cognifyz Machine Learning Internship

Task: Cuisine Classification

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1. Objective

The aim of this task is to build a machine learning model that can **predict the cuisine type of a restaurant** using the information available in the dataset. The model learns patterns from restaurant features such as cost, location, ratings, and delivery options.

2. Dataset

- **Dataset Name:** Restaurant.csv
- **Source:** Cognifyz Technologies

The dataset contains information about **9,551 restaurants**, including their names, locations, prices, ratings, and cuisine types.

The **cuisine** column is used as the target variable.

3. Method Used

- The dataset was loaded and cleaned
- Missing values were handled
- Categorical data was converted into numbers using **Label Encoding**
- Basic Exploratory Data Analysis (EDA) was performed to understand cuisine distribution
- Important features such as cost, votes, ratings, delivery, and location were selected

4. Model Building

A **Random Forest Classifier** was used to predict cuisines because it works well with multi-class and complex data.

Steps followed:

- Data was split into training (80%) and testing (20%) sets
- The model was trained using 200 decision trees
- Predictions were made on test data

5. Evaluation

- Accuracy achieved was around **18%**
- Lower accuracy is expected due to **high imbalance** in cuisine categories
- Common cuisines were predicted better than rare ones

Despite this, the model successfully performs multi-class cuisine classification.

6. Conclusion

The cuisine classification model was successfully implemented using Random Forest.

Although performance is limited by data imbalance, the project demonstrates a complete machine learning workflow and meets the internship task requirements.

7. Future Improvements

- Balance cuisine classes using SMOTE
- Try advanced models like XGBoost or SVM
- Group rare cuisines into a single category
- Build a web interface using Streamlit or Flask